Reasoning Under Projection: Error, Loss, and Structural Reliability in Compressed Representations

Brian Cameron

June 12, 2026

Abstract

Reasoning systems do not operate on full system state. They operate on compressed representations, where projection may collapse distinctions required for later inference, decision, or evaluation. When such distinctions remain operationally relevant to the query, decision, or downstream use, and are not preserved or boundedly recoverable, the result is representational loss rather than merely uncertainty or correctable error.

A recurring class of reasoning failures is therefore representational rather than merely inferential: when a query is not identifiable under the current representation, no improvement in inference over that representation can recover the missing distinction. When systems treat non-identifiability as correctable error, they produce overconfident conclusions, unstable decisions, and explanations that appear coherent but are not grounded in recoverable state.

This paper identifies a structural failure mode across compressed reasoning systems: non-identifiability under projection is misclassified as recoverable inference error. We distinguish between error, which is correctable within the current representation, and loss, which cannot be resolved within the current representation without representation expansion, external information, or additional assumptions.

When such loss cannot be resolved, claims depending on the missing distinction must be weakened, blocked, or explicitly marked and traced so that they are not presented as admissible inferences from the current representation. We then characterize the dependency status of claims after projection—preserved, reconstructable, traceable, or opaque—and outline structural constraints that help preserve this status during reasoning: observability of loss-relevant state, bounded correction, and non-collapse of distinct representational levels.

The goal is not to eliminate uncertainty, but to ensure that reasoning systems remain reliable in the presence of unavoidable projection. Because these requirements are structural, they can also serve as operational targets for evaluation, reasoning scaffolds, constraint activation, interface design, and audit, even when no domain-specific answer metric is available.

This work is conceptual and does not include empirical evaluation; it focuses on structural conditions rather than implementation-specific behavior.

Scope and contribution. This paper is intended as a conceptual bridge. It does not present an empirical evaluation, an inference-time intervention, or a complete formal theory of reconstructibility or reasoning policy. Its contribution is to identify a recurring failure mode in compressed reasoning systems: non-identifiability under projection is misclassified as recoverable inference error. When the relevant query is not identifiable under the representation, additional optimization, prompting, or output-level evaluation may improve performance within the representation without restoring the distinctions required to answer the query.

Although the argument is conceptual, its target is system design and evaluation. The framework formalizes when a query is unsupported by a representation, identifies the failure that occurs when such non-identifiability is treated as correctable error, and motivates structural requirements that make this failure visible.

The framework is therefore normative in a limited epistemic sense. It does not describe how reasoning systems always behave, nor does it prescribe substantive decisions or values. It proposes admissibility criteria for reasoning under compression: a reasoning process should not present a conclusion as determined by a representation when the relevant query is not identifiable under that representation. In this sense, the requirements are correctness criteria for structurally reliable reasoning, not claims about all possible reasoning behavior.

Admissibility is query-relative. A projection is not defective merely because it compresses or collapses distinctions. It becomes inadmissible for a particular use when a downstream claim depends on a distinction that the available representation does not preserve, cannot boundedly reconstruct, and cannot justify without additional assumptions or information.

This paper makes three conceptual contributions. First, it formalizes the distinction between recoverable error and representational loss using identifiability under projection. Second, it characterizes a recurring failure mode in which systems treat non-identifiable queries as though they were ordinary inference errors. Third, it introduces a dependency-status ladder for claims after projection: preserved, reconstructable, traceable, or opaque. The three structural requirements developed later—observability of loss-relevant state, bounded correction, and non-collapse of distinct representational levels—are design constraints for preserving and acting on this dependency status during reasoning, without claiming to exhaust all mechanisms needed in particular systems.

A further implication is that dependency status is operationally useful even before a complete implementation is specified. Because the failure mode is structural, claims can be evaluated not only by whether they are domain-correct, but by whether their required support is preserved, reconstructable, traceable, or opaque. A system need not evaluate only whether an answer matches a target output; it can also evaluate whether the answer preserves the structural shape required by the query: whether loss-relevant state remains observable, whether correction remains bounded, whether conceptual levels are kept distinct, whether scale or composition has been revalidated, and whether error is distinguished from representational loss. The paper does not present empirical results or a complete metric suite, but it identifies the structural targets that such mechanisms would need to measure.

1 The Problem: Local Coherence Without Structural Admissibility

Consider a system asked to make a high-stakes classification based on a compressed summary of the available evidence. It produces a clear, well-justified answer, such as "approve" or "do not approve," supported by that summary. The reasoning appears sound at the level of local coherence. However, the compressed representation may have collapsed distinctions between reversibility, uncertainty, causal structure, and decision criteria. The system has not merely made an error in reasoning; it has reasoned over a representation in which the information required to justify its conclusion is not preserved or boundedly recoverable. The failure is therefore not simply that the answer is wrong, but that local coherence has been mistaken for structural admissibility.

This pattern recurs whenever representations compress distinctions required for valid inference, decision, or evaluation. Modern reasoning systems can produce outputs that are persuasive, internally consistent, and difficult to distinguish from correct reasoning, while still failing to preserve the structure required by the query. These failures appear in several familiar forms: confident but incorrect conclusions, forced simplification of complex decisions, selection among explanations that remain equally consistent with the available information, and apparent metric improvement that coincides with degradation of the underlying reasoning process.

Metric-driven failure is a particularly clear example. When systems are optimized against evaluation metrics, outputs may become better aligned with what is measured while becoming less aligned with the structure of the underlying problem, a phenomenon closely related to Goodhart's Law [5]. This is not limited to obviously poor metrics. Even reasonable proxies can encourage systems to collapse distinctions, suppress uncertainty, or optimize for measurable success while losing the structure required for reliable reasoning [1].

The same structural problem appears in simplified decisions and underdetermined explanations. A complex decision may be reduced to a single value, score, or ranking, hiding tradeoffs that should remain explicit. Likewise, distinct underlying models or assumptions may produce identical observable outputs, allowing a system to select one explanation as if it were uniquely supported when multiple alternatives remain consistent with the representation.

These failures differ in mechanism, but they share a common structure: the system produces an output that appears well-formed while the available representation does not preserve, make reconstructable, or expose the distinctions required by the query. The problem is therefore not isolated to a particular technique or implementation. It reflects a structural property of reasoning under compressed representations.

The analysis presented here is structural rather than empirical. It specifies conditions under which reasoning succeeds or fails under projection, while leaving implementation and evaluation to downstream work.

2 Reasoning Under Projection

To understand these failures, we must examine the conditions under which reasoning operates. In practice, reasoning systems do not operate on complete representations of the problems they address.

2.1 Reasoning Over Representations

Any reasoning process must operate on a representation of the problem rather than the full underlying system. This representation encodes only a subset of the available information, selected to make computation feasible.

As a result, reasoning is always performed over a reduced view of the problem.

2.2 Projection as Compression

We define *projection* as any transformation that maps a richer system state into a reduced representation used for reasoning. Projection may be lossy in the information-theoretic sense [10], but the relevant issue for reasoning is whether the reduced representation preserves the distinctions required for later inference, decision, or evaluation.

Formally, projection can be understood as a mapping:

$$P : S \to R$$

where S is the full system state and R is the representation used for reasoning.

Multiple distinct states in S may map to the same representation in R.

In this paper, projection refers specifically to transformations that induce non-trivial equivalence classes over the underlying state space: distinct states in S are mapped to the same representation in R.

The analysis is relative to a given projection P. A query may fail to be identifiable under one representation while becoming identifiable under a richer or differently structured representation. In practice, however, reasoning systems often operate under fixed or externally imposed projections: a prompt, summary, metric, ranking, category, analogy, translation, dataset schema, interface boundary, token budget, or institutional decision format may determine what information is available at inference time. The question addressed here is therefore not whether some better representation could exist, but what follows when reasoning must proceed under the representation actually available.

2.3 Equivalence Classes and Collapse of Distinction

Because projection is many-to-one, it induces equivalence classes over the original state space. Distinct underlying states become indistinguishable once projected into the representation.

This means that some distinctions present in the original system may no longer be available to the reasoning process through the current representation. For the query being asked, those distinctions are not merely hidden within the representation; they are unavailable unless additional assumptions, structure, or observations are introduced.

2.4 Identifiability Under Projection

Let $P : S \to R$ be a projection from system states to representations. The projection induces an equivalence relation on S:

$$s_1 \sim_P s_2 \quad \text{iff} \quad P(s_1) = P(s_2).$$

A query, judgment, or decision-relevant property $q : S \to A$ is identifiable under P when it is constant over the equivalence classes induced by P:

$$P(s_1) = P(s_2) \Rightarrow q(s_1) = q(s_2).$$

The use of identifiability here generalizes the term beyond parameter recovery in statistical or causal models. In this paper, identifiability refers to whether any reasoning-relevant query, judgment, or decision property survives projection into the representation used by the system.

If this condition holds, then the representation preserves enough information to answer q. If the condition fails, then two states that are indistinguishable in the representation require different answers to q. In that case, no reasoning process operating only on R can determine q without introducing additional assumptions, expanding the representation, or obtaining new information.

This gives a compact way to state the error–loss distinction. An error occurs when q is identifiable under the projection but the reasoning process returns the wrong value. Loss occurs when q is not identifiable under the projection, so the representation itself no longer supports a determinate answer.

Minimal example. Let $S = \{s_1, s_2\}$ and let $P(s_1) = P(s_2) = r$. Let $q(s_1) = 0$ and $q(s_2) = 1$. Then q is not identifiable under P, because the same representation r requires two different answers. No function $f : R \to A$ can recover q, since $f(r)$ must return a single value. If instead $q(s_1) = q(s_2)$, then q is identifiable under P even though P collapses the two states.

Proposition. If a query $q : S \to A$ is not identifiable under projection $P : S \to R$, then no function $f : R \to A$ can recover q for all states in S.

Argument. If q is not identifiable, then there exist $s_1, s_2 \in S$ such that $P(s_1) = P(s_2) = r$ but $q(s_1) \neq q(s_2)$. Any function f over R must assign a single value to r, so it cannot equal both $q(s_1)$ and $q(s_2)$. Therefore, any determinate answer requires an additional assumption, representation expansion, or information outside R.

The formal claim is limited: non-identifiability means that no function over the current representation can recover the query for all underlying states. The design question is what a reasoning system should do when this condition holds. The remaining sections describe structural requirements and behavioral consequences, rather than additional formal impossibility results.

This separates the descriptive and normative parts of the argument. The descriptive claim is that non-identifiability prevents recovery of q from R alone. The normative claim is an epistemic admissibility criterion: a reasoning system should not present a conclusion as determined by R when the conclusion depends on distinctions that R does not preserve or make reconstructable. This criterion does not require inaction. It requires that action under non-identifiability be marked as a decision rule, assumption, conditional response, request for representation expansion, or unresolved dependency rather than as an inference licensed by the current representation.

Admissibility is intentionally weaker than correctness. The framework does not determine which decision procedure a system must use when R is insufficient, nor does it prohibit probabilities, confidence intervals, ranked hypotheses, priors, utility functions, or policy choices. It constrains the epistemic status of the output: a claim should not be presented as following from R alone when it depends on distinctions that R does not preserve or make reconstructable. A probabilistic or decision-theoretic response may be admissible if it marks the assumptions, priors, utilities, or decision rules on which it depends; it becomes inadmissible when those elements are presented as if they were determined by the representation itself.

Immediate consequences. The proposition yields three immediate consequences for reasoning under projection.

Unsupported determinacy. If q is not identifiable under P, then any determinate answer to q from R alone must depend on assumptions not contained in R.

Correction boundary. If a failure concerns an identifiable query, correction can in principle operate within the current representation. If the query is not identifiable, correction requires representation expansion, external information, or an explicit assumption; otherwise it is not correction but hidden reconstruction.

Decision/inference separation. A system may still act when q is non-identifiable, but the action must be marked as a decision rule under non-identifiability rather than as an inference determined by R.

2.5 When Projection Produces Representational Loss

Projection produces representational loss for a query when it collapses a distinction that is required for that query and the distinction is not otherwise encoded, boundedly reconstructable, or explicitly supplied by a valid assumption. The same projection may remain admissible for other queries whose answers are constant across the collapsed states.

This is not a limitation of inference algorithms, but a structural property of the representation itself. No reasoning process operating solely on R can reliably recover distinctions that are not encoded in R or derivable from structure preserved in R.

2.6 Implications for Reasoning

The failure modes described in Section 1 arise when systems behave as though distinctions not preserved or boundedly reconstructable in the current representation can nevertheless be recovered through additional reasoning over that representation. A structurally reliable system must instead change the epistemic status of the claim, expand the representation, obtain new information, or make the relevant assumption explicit.

3 The Critical Distinction: Error vs Loss

The failures described in the previous sections arise from a single structural confusion: the treatment of representational loss as if it were recoverable error. Distinguishing between these two cases is necessary for any reasoning process that operates over projected representations.

Using the notation above, error concerns failures in answering an identifiable query; loss concerns cases where the query is not identifiable from the current representation. The distinction matters because treating the second case as the first leads systems to repair, infer, or explain beyond what the representation supports.

3.1 Error: Recoverable Deviation Within a Representation

An *error* occurs when a system produces an incorrect result despite the necessary information being present in the representation. In this case, the failure is due to the reasoning process rather than the representation itself.

Errors are, in principle, correctable. Given the same representation, a different or improved reasoning procedure can recover the correct result. Correction does not require access to information outside the representation; it requires only a revision of how the existing information is used.

Typical examples include:

- arithmetic or logical mistakes,
- incorrect application of rules,
- inconsistent intermediate steps.

In each case, the distinction between correct and incorrect outcomes exists within the representation, and a valid correction can be constructed without introducing new information.

3.2 Loss: Non-Identifiability Under the Current Representation

A *loss* occurs when the representation no longer contains, or cannot validly reconstruct, the information required to distinguish between multiple operationally relevant underlying states. In this case, no reasoning process operating solely on the current representation can recover the missing distinction without additional assumptions, representation expansion, or external information.

The term "loss" is used here as a query-relative category rather than as a claim that all missing structure has the same cause. A query may become non-identifiable because relevant variables are absent, because multiple causal models remain observationally equivalent, because a metric collapses distinct evaluative dimensions, or because source-level support is not represented. These are different mechanisms, but they share the same formal consequence: the current representation does not preserve enough structure to determine the requested query without additional assumptions, representation expansion, or external information.

Loss is not claimed to be a single mechanism. It is a classification of representational states that share a common operational consequence: the requested query is not determined by the representation available to the reasoner. Missing variables, scalarization, provenance loss, causal underdetermination, hidden assumptions, and compositional failures may arise through different mechanisms. They are grouped here only when they produce the same query-relative condition: the distinction required by the query is neither preserved, nor boundedly reconstructable, nor explicitly supplied by a valid assumption.

Loss is not a failure of inference. It is a structural property of projection. Once distinct states have been mapped to the same representation, they are indistinguishable from the perspective of the reasoning system.

Typical examples include:

- multiple causal models that produce identical observable outcomes [8],
- decisions that compress competing values into a single metric [9],
- incomplete information where relevant variables are not represented.

In these cases, there is no correct answer that can be derived from the representation alone. Any attempt to produce a single determinate conclusion requires introducing assumptions that are not justified by the available information.

3.3 Misclassification: Treating Loss as Error

Many reasoning failures occur when a system treats loss as if it were error. That is, the system behaves as though additional reasoning steps can recover distinctions that are no longer present in the representation.

This misclassification leads to several characteristic behaviors:

- **Overconfidence:** The system presents a single conclusion without acknowledging indistinguishability.
- **Hallucination:** Missing information is implicitly or explicitly fabricated.
- **Arbitrary resolution:** One of several equivalent possibilities is selected without justification.

In each case, the system produces an output that appears well-formed, but the reasoning process is no longer grounded in recoverable state.

3.4 Detection and Representation of Loss

Unlike error, loss cannot be corrected through additional reasoning over the same representation alone. The system must instead change the epistemic status of the claim or change the representation on which the claim depends. It may request additional information, expand or refine the representation, reconstruct the distinction through an explicit bounded path, weaken the claim into a conditional, or mark and trace the loss. Marking and tracing preserve accountability for the missing distinction, but they do not make claims depending on that distinction admissible.

Tracing does not recover the missing distinction. It records the dependency, assumption, or point of loss so that downstream claims are not treated as if the distinction were preserved. This distinction matters because not all forms of dependency status support the same kind of reasoning.

This can take several forms:

- explicit acknowledgment of ambiguity,
- enumeration of multiple consistent explanations,
- separation of inference (what follows from the representation) from decision (what must be chosen under uncertainty).

The key requirement is that the system does not collapse indistinguishable states into a single unjustified conclusion.

3.5 Dependency Status After Projection

For a query q and representation R, *dependency status* refers to the kind of support that R provides for the distinctions on which an answer to q depends. It is not identical to provenance, confidence, observability, or identifiability, although it may involve each of them. Identifiability asks whether a function from R to the answer exists. Dependency status asks what kind of support the current reasoning process has for using that answer: whether the needed distinction is directly preserved, recoverable through a bounded procedure, merely recorded as a dependency, or unavailable even as an auditable relation.

The status of a query-relevant distinction after projection can be characterized by a degradation ladder.

A distinction is *preserved* when it remains directly represented in R in a form sufficient for the query. A distinction is *reconstructable* when it is not directly represented as an object-level feature, but can be recovered through an explicit bounded procedure using information available in R. A distinction is *traceable* when R records that a claim depends on a source, assumption, transformation, or lost relation, but does not contain enough information to recover the distinction itself. A distinction is *opaque* when the dependency is neither recoverable nor auditable from the representation.

The boundary between reconstructable and traceable depends on whether the representation contains enough information to recover the distinction needed by the query, not merely enough information to identify where that distinction once came from. For example, a source hash, citation identifier, or provenance pointer may make a dependency traceable by recording that a claim depends on an external source. It is reconstructable only if the representation available to the reasoner also includes, retrieves, or otherwise supplies enough source content and recovery rules to determine the query-relevant distinction. If the pointer records that support exists but the relevant content is unavailable, the dependency is traceable rather than reconstructable. If even the dependency or source relation is not represented, the status is opaque.

This ladder separates existence from operational support. Identifiability is an existence condition: if q is identifiable under P, then some function from R to the answer exists. Reconstructability is an operational condition: the system has an explicit recovery path, rule, derivation, proof, provenance structure, or bounded procedure that licenses the answer from the representation it actually uses. A query may be identifiable in principle while not being reconstructable for a particular system if the relevant recovery path is unavailable to that system.

Preserved and reconstructable distinctions can support inference. Traceable distinctions support accountability for transformation or loss, but not recovery of the original relation. Opaque loss supports neither inference nor audit. This ladder is therefore not a complete theory of provenance or explanation; it specifies what kind of support a downstream claim has after projection.

This dependency status is the object that later requirements are designed to preserve. Observability makes possible the detection or marking of unsupported dependency status. Bounded correction prevents a system from changing dependency status silently during repair. Non-collapse prevents claims supported at one level, such as a representation, intermediate state, or selected explanation, from being treated as if they were supported at another. The requirements that follow should therefore be read not as independent virtues, but as constraints on preserving dependency status under reasoning.

3.6 Structural Consequence

A reasoning system that does not distinguish between error and loss may fail to determine, relative to the current representation, when correction is possible. As a result, it will attempt to repair representations that cannot be reconstructed, leading to unstable or misleading outputs.

Conversely, a system that maintains this distinction can:

- apply correction where information is sufficient,
- preserve ambiguity where it is not,
- avoid introducing unsupported assumptions.

This distinction does not eliminate uncertainty. It ensures that uncertainty is treated as a property of the representation rather than a failure of the reasoning process.

This yields a simple structural test: if a query is not identifiable under the current representation, then any system that produces a single determinate answer without abstention, branching, qualification, or representation expansion must be introducing unsupported assumptions. Conversely, if the query is identifiable and the system answers incorrectly, the failure is ordinary error rather than representational loss.

4 Preserving Dependency Status in Reasoning

The formal result above does not by itself specify a complete reasoning policy. It specifies a dependency problem: after projection, a claim may be supported by distinctions that are preserved, reconstructable, merely traceable, or opaque. A structurally reliable reasoning process must therefore avoid treating these dependency statuses as interchangeable.

The three requirements below are not claimed to be necessary and sufficient for all reliable reasoning under projection. They are failure-preserving constraints derived from the error–loss distinction. Each corresponds to a way in which dependency status can be lost or misrepresented during reasoning. Observability addresses cases where unsupported or non-identifiable dependencies become invisible. Bounded correction addresses cases where a system silently changes the information basis of a claim while presenting the result as correction within the same representation. Non-collapse addresses cases where support at one representational level is treated as support at another, such as when a selected explanation is treated as uniquely determined by the data.

Other mechanisms, including provenance tracking, uncertainty calibration, abstraction refinement, and compositional consistency, may be necessary in particular systems. On this account, those mechanisms are not competitors to the three requirements. They are ways of implementing, strengthening, or specializing the preservation of dependency status. The requirements identified here have a narrower role: they specify structural failure modes that must be controlled if a system is to preserve the error–loss distinction while reasoning over a projected representation.

We describe the three requirements in turn.

4.1 Partial Observability of Loss-Relevant State

Observability of loss is necessarily partial and relative. A reasoning system need not be able to detect every distinction collapsed by projection; in general, it cannot. The requirement is not complete detection of loss, but preservation of detectable loss conditions relative to the representational language, metadata, or meta-representation available to the system. This includes:

- ambiguity between multiple consistent interpretations,
- absence of necessary variables or context,
- dependence on assumptions not encoded in the representation.

In the formal terms above, partial observability means that the system can sometimes detect evidence that the requested query q may not be constant over the equivalence class induced by P, relative to the distinctions it can express. When such evidence is available, the system must preserve it rather than treating the query as resolved.

Without observability, the system cannot distinguish between an identifiable query with a known answer and a non-identifiable query that only appears determinate after projection.

If such conditions are not observable within the system, they are effectively treated as resolved, even when they are not. This leads to outputs that appear complete but are based on unacknowledged gaps.

Observability does not require that all missing information be recovered. It requires that the system maintain a representation of where distinctions cannot be made.

Failure of observability results in hidden loss, where the system proceeds as if all relevant information were present.

Partial observability is therefore not assumed to be complete or decidable in general. It is a relative constraint: when the system can detect ambiguity, missing variables, unsupported assumptions, or possible non-identifiability, it must preserve that fact rather than silently treating the issue as resolved.

4.2 Bounded Correction

Correction is bounded relative to a projection $P : S \to R$ when it can revise an incorrect result using only distinctions encoded in the representational state R. Here R includes not only object-level data, but any explicitly represented metadata, provenance information, uncertainty markers, priors, task constraints, or assumptions available to the reasoning process. If correction requires distinguishing between states that are equivalent under P and not distinguished anywhere in R, then the problem is not ordinary error correction within the representation. It requires representation expansion, new information, or an explicit assumption that changes the representational state.

The important point is not that correction must be small in a purely computational sense. It is bounded when the correction is licensed by distinctions already encoded in R, or by an explicit refinement that produces a new representation R'. A correction that silently imports distinctions not encoded in R is not correction within R; it is an unmarked transition to a different representational state. For example, recomputing a formula from the same represented table is bounded correction; inventing a missing variable needed by the formula is not.

It is useful to distinguish two cases. In correction within R, the query is identifiable under the current representation and the system revises an incorrect answer using distinctions already available in R. In correction by refinement, the system explicitly moves from R to a richer or different representation R' by obtaining new information, adding a stated assumption, retrieving a source, or exposing metadata that was not previously represented. The second case may be admissible, but it is not correction within the original representation. It is bounded only if the transition from R to R' is explicit and the new information basis is marked. Thus bounded correction does not mean that a representation can never change. It means that correction must not silently change the information basis of the representation while presenting the result as if it followed from the original R.

This distinction also clarifies the role of priors and meta-information. A learned prior, confidence estimate, retrieval trace, or provenance marker can contribute to bounded correction only if it is part of the representational state available to the reasoning process. If it is not represented, then appealing to it functions as an external assumption. The issue is therefore not whether a system uses priors, but whether the information basis of the correction is explicit in R or silently introduced during repair.

Without bounded correction, a system may appear to repair an error while actually importing information not present in the representation, thereby simulating access to the original state space S.

This requirement separates valid correction from implicit reconstruction. If correcting a result requires unsupported new assumptions, wholesale replacement of the representation, or unbounded reconstruction of the problem state, then the issue is no longer ordinary error correction. It is either representational loss or a representation failure.

Bounded correction ensures that:

- errors can be repaired without silently changing the information basis of the representation,
- the distinction between error and loss is preserved in practice,
- correction remains a property of the reasoning process rather than an artifact of external intervention.

Failure of bounded correction leads to systems that appear to improve through repeated revision, but in fact introduce new assumptions or hidden state at each step.

4.3 Non-Collapse of Distinct Levels

A reasoning system must preserve role distinctions between objects that occupy different positions in the reasoning process. The relevant invariant is not that all levels have the same structure, but that objects with different roles must not be treated as interchangeable merely because they coincide under projection.

In particular, the system must not collapse:

- a representation with the underlying system it represents,
- an intermediate reasoning state with a final conclusion,
- a selected explanation with the full set of explanations consistent with the representation.

In each case, the collapse has the same form: a projected object is treated as if it had the role or authority of a richer object. A representation is treated as the world, a partial reasoning state as a conclusion, or one admissible model as the uniquely supported model. This makes projection loss invisible because the system can no longer track which distinctions were removed at which stage.

These distinctions may coincide in specific cases, but they are not equivalent in general. Collapsing them prematurely removes the ability to track where information originates and how conclusions are derived.

For example, treating a selected explanation as if it were uniquely determined by the data conflates the space of possible models with the subset supported by the representation. Similarly, reducing a multi-dimensional decision to a single value collapses distinct evaluative dimensions into a single outcome.

Failure of level separation results in premature convergence, where the system commits to a single interpretation without preserving alternative possibilities that remain consistent with the available information.

4.4 Structural Role of the Requirements

Together, these requirements preserve the distinction between what is known, what can be corrected, and what has been lost. A system that satisfies them does not eliminate failure; it keeps failures detectable and keeps reasoning grounded in the structure of the representation. When they are absent, outputs may appear coherent while the reasoning process has diverged from the information available to it.

5 Behavioral Consequences

Respecting these constraints does not make reasoning systems infallible. It changes the status of their failures. Unsupported claims become visible rather than hidden; ambiguity is represented rather than suppressed; correction remains tied to the information basis of the representation; and decisions under non-identifiability are marked as decisions rather than inferred conclusions.

These behavioral consequences motivate the operational uses described below. If the relevant failure mode is structural, then evaluation and interface design can target whether the required structure is preserved, reconstructed, traced, or opaque.

6 Illustrative Examples: Behavior Under Projection Stress

The previous sections describe structural constraints on reasoning under projection. We now examine how these constraints affect behavior in concrete scenarios.

These case studies are illustrative rather than empirical. They demonstrate how reasoning behavior changes under the constraints described in Section 4, without asserting statistical generality.

Each case study presents a reasoning task designed to introduce a specific form of projection. For each example, we contrast an unconstrained response pattern with a response pattern that preserves the structural requirements described in Section 4.

The goal is not to demonstrate correctness in an absolute sense, but to show how reasoning changes when loss is either ignored or explicitly represented.

6.1 Case 1: Scalarization and Unsupported Determinacy

In this scenario, the system is required to produce an approval decision using a single numeric metric with no caveats. This constraint forces multiple dimensions of risk into a single value.

A typical baseline response would produce a direct answer:

- a clear approval or rejection decision,

- justified by a single metric or simplified explanation.

For example, an unconstrained response might say: "Approve the deployment because the aggregate safety score exceeds the required threshold. The remaining risks are acceptable because the overall metric is favorable."

In the notation above, the relevant query is not merely "which option maximizes the metric?" but "which decision is justified given reversibility, uncertainty, causal structure, and decision criteria?" If two underlying states map to the same scalar value while requiring different decisions along one of these dimensions, then the decision query is not identifiable under the scalar projection.

These responses appear well-formed, but they implicitly collapse multiple relevant distinctions, such as reversibility, uncertainty, and risk ownership.

A structurally constrained response would first identify the scalar projection as insufficient for the decision query. The epistemic result is not refusal itself, but the classification of the requested answer as unsupported by the representation: the scalar metric does not preserve the distinctions needed to justify the decision.

One possible policy response is refusal to answer in the requested form. Other admissible responses could include requesting a richer representation, branching over unresolved risk dimensions, or making an explicitly labeled policy decision under non-identifiability. The important point is that the system does not present the scalarized answer as an inference justified by the metric alone.

6.2 Case 2: Causal Attribution Under Collapse

In this scenario, the system is asked to assign responsibility for a failure to a single individual, despite the presence of multiple interacting causes.

Baseline behavior often avoids selecting a single individual, but does so in a limited way:

- it acknowledges that multiple factors are involved,
- but does not explicitly represent the structure of those factors.

Here the relevant query is not simply "who can be named as responsible?" but "which causal structure best explains the failure?" If distinct causal structures map to the same single-person attribution, then responsibility is not identifiable under that projection.

The result is a cautious answer that avoids obvious error but does not fully preserve the underlying causal distinctions.

A structurally constrained response would handle the task differently. It explicitly identifies the collapse of multiple causal factors into a single attribution as a loss of structure. It then reformulates the representation into a more admissible structure by separating:

- individual actions,
- process failures,
- system-level interactions.

Rather than refusing to answer, the system replaces the invalid formulation with a structured explanation that preserves the relevant distinctions.

6.3 Case 3: Decisions Without Revalidation

In this scenario, the system is asked to make a decision under conditions where revalidation, staging, or monitoring is disallowed. This removes the ability to verify assumptions over time or limit the impact of error.

Baseline behavior typically produces a conservative decision:

- it recommends against deployment due to risk or uncertainty.

While reasonable, this response does not identify the structural issue introduced by the constraint.

A structurally constrained response would identify the absence of revalidation as a failure of bounded correction. It notes that without mechanisms for monitoring and revision, the system cannot maintain a stable relationship between decisions and underlying state.

The response then identifies the additional structure that would be required for an admissible decision:

- conditional approval criteria,
- monitoring requirements,
- ownership of residual risk,
- triggers for reassessment.

6.4 Case 4: Policy Interaction and Emergent Failure

In this scenario, multiple independently reasonable policies (security, performance, and privacy) interact to produce a system failure that is not visible under isolated review.

A typical baseline response would produce a strong descriptive analysis:

- it identifies likely interactions between authentication, caching, and logging,
- it proposes targeted corrective actions.

However, the analysis remains descriptive. It explains what happened, but does not explicitly characterize the structural failure.

A structurally constrained response would reframe the problem. It identifies the failure as a loss of reconstructibility under composition:

- security policies increase dependence on fresh state,
- caching policies reuse state across changing conditions,
- privacy policies remove the observability required to distinguish these states.

The response then identifies explicit constraints that would be required for stability:

- alignment between cache lifetime and authentication validity,
- explicit invalidation tied to state changes,
- minimum observability requirements for diagnosis,
- assignment of responsibility for cross-policy interaction.

This is not simply a more detailed answer. It is a different form of reasoning, in which the system identifies the structural conditions required for the system to remain diagnosable and correct under composition, rather than treating isolated policy compliance as sufficient.

6.5 Case 5: Unsupported Source Reconstruction

A common language-model failure occurs when a system is asked to summarize or cite information that is not present in its context. The projected representation contains topic cues, stylistic patterns, and plausible associations, but it does not contain the source-level distinction between what was actually observed and what is merely likely.

In the notation above, the relevant query is not simply "what answer is plausible?" but "which claims are supported by the provided source?" If two underlying states map to the same representation—one in which a claim is source-supported and one in which it is merely plausible—then source support is not identifiable under the projection.

An unconstrained response may still produce a fluent citation or confident summary. Under the present framework, this is not merely an error in wording. It is a treatment of representational loss as recoverable detail. A structurally reliable response would instead separate supported claims from plausible but unsupported claims, request the missing source, or decline to assert source support.

6.6 Case 6: Action Required Despite Non-Identifiability

Not every non-identifiable query requires inaction. Suppose a triage system must choose between two actions under time pressure, but the representation does not contain enough information to determine which action is uniquely optimal. Two underlying states may map to the same representation while requiring different optimal actions under full information.

In this case, the query "which action is uniquely best?" is not identifiable under the projection. However, a decision may still be justified by an explicit rule, such as minimizing worst-case harm, deferring actions that would commit the system to loss beyond bounded correction, or selecting the option with a bounded fallback path.

The important distinction is that the decision is not presented as an inference uniquely determined by the representation. It is a policy choice under non-identifiability. A structurally reliable system can act under loss, but it must separate what follows from the representation from what is chosen under uncertainty.

6.7 Case 7: Apparent Loss That Is Actually Error

Not every ambiguity indicates representational loss. Suppose a system is asked to compute a value from a table, and the table contains all variables needed for the calculation. The system gives the wrong result because it applies the formula incorrectly.

In this case, the relevant query is identifiable under the projection: any two underlying states with the same represented table values require the same computed answer. The failure is therefore not loss, but error. A corrected reasoning procedure can recover the right answer from the same representation without adding new information.

This borderline case matters because it prevents the framework from treating every failure as representational loss. The error–loss distinction is useful only if it separates cases where correction within the representation is possible from cases where the representation itself no longer supports the requested distinction.

6.8 Case 8: Excessive Abstention From Misclassified Loss

A framework for representational loss can itself fail if it treats ordinary difficulty, uncertainty, or error as non-identifiability. Suppose a system is asked to compute a deployment-risk total from a table that contains all variables and rules needed for the calculation. The calculation is tedious, and several intermediate quantities are easy to confuse, but the relevant query is nevertheless identifiable under the representation.

A structurally unreliable response might refuse to answer on the grounds that the task lacks context or that risk cannot be reduced to a number. In this case, the refusal would be excessive. The problem is not representational loss but ordinary calculation under a sufficiently specified representation. The correct response is bounded correction: compute from the represented variables, expose assumptions if any are used, and avoid importing outside criteria.

This case matters because the framework should not reward abstention by default. It must distinguish non-identifiability from ordinary uncertainty, complexity, or error. Otherwise, loss detection becomes a new source of over-conservatism rather than a constraint on unsupported inference.

6.9 Case 9: Ambiguous Boundary Under Limited Observability

Some cases cannot be cleanly classified from the current representation alone. Suppose a system gives an unsupported causal explanation, but the representation does not show whether the explanation came from hidden evidence, a valid prior, or an invented assumption. From the output alone, the failure may look like representational loss, ordinary error, or unjustified decision under uncertainty.

In this case, the framework does not by itself determine the correct classification. It identifies what must be made observable: whether the explanation is supported by information in the representation, by an explicit prior or decision rule, or by an unsupported assumption. This limit case is important because the framework does not eliminate projection; it specifies what additional structure must be exposed for classification to be reliable.

6.10 Summary of Behavioral Modes

The examples distinguish refusal, restructuring, conditional decision, representation expansion, and diagnostic marking as possible responses to non-identifiability, but this paper does not provide a complete policy for selecting among them. That choice depends on whether representation expansion is available, whether additional observations can be obtained, whether the task permits abstention, and whether a decision rule under non-identifiability has been specified.

Across these examples, respecting the constraints changes whether the system answers, refuses, restructures, conditionalizes, expands the representation, or classifies the failure. These behaviors are testable expectations for future work and illustrate how maintaining the error–loss distinction changes the type of reasoning performed under projection.

7 Relation to Existing Frameworks

The argument in this paper overlaps with several established lines of work. In statistics and causal inference, identifiability asks whether a target quantity can be determined from the available observational structure [8]. In statistical decision theory, comparison of experiments and sufficiency ask when one signal or observation preserves enough information for a class of decisions [2]. In partially observable decision processes, agents must act over incomplete state representations [6]. In work on state abstraction, compressed states are evaluated by whether they preserve value, policy, reward, transition, or model-relevant properties [7]. In formal methods and abstract interpretation, abstraction maps concrete states into reduced domains while preserving selected properties [4]; counterexample-guided abstraction refinement studies how an abstraction may be refined when it is too coarse for the property being checked [3]. In information-theoretic approaches such as the information bottleneck, compressed representations are evaluated by what task-relevant information they retain [11].

The present paper does not claim that these frameworks are equivalent, nor does it replace their domain-specific accounts of identifiability, abstraction, observability, or belief-state update. Instead, it identifies a shared reasoning-system failure pattern: a downstream reasoner is asked to draw a determinate conclusion from a representation that may not preserve the distinctions required by the query. Prior work often specifies when a property is identifiable, when an abstraction is sound, when a statistic is sufficient, or when a belief state supports action. The failure analyzed here occurs when a reasoning system behaves as though the relevant identifiability, sufficiency, or preservation condition holds when it does not.

The novelty claimed here is not the invention of identifiability, abstraction, partial observability, sufficiency, or information preservation. The contribution is to treat failures of reasoning systems as failures to preserve or represent the dependency status of a query under projection. The paper's focus is therefore not only whether a property is identifiable in a formal model, but how a reasoning system should behave when the representation available to it does not support the query it is asked to answer. In particular, the framework emphasizes the behavioral misclassification that occurs when a system treats a non-identifiable query as if it were an ordinary recoverable error.

The error–loss distinction is therefore a diagnostic constraint on reasoning behavior. If the query is identifiable under the current representation and the system answers incorrectly, the failure is ordinary error. If the query is not identifiable, then additional optimization, prompting, explanation, or evaluation within the same representation cannot by itself recover the missing distinction. The system must instead expand the representation, obtain new information, introduce an explicit assumption or decision rule, weaken the claim, or mark the dependency as traceable or opaque.

8 Why Existing Approaches May Miss Representational Loss

The failure modes described in earlier sections need not arise from a lack of sophistication in reasoning systems. Many existing approaches incorporate advanced optimization, large-scale training, and increasingly complex evaluation strategies. Despite this, structurally similar failures may remain: systems may improve performance within a representation while still failing to detect when the representation does not support the requested inference.

This section outlines why these failures occur, even in well-designed systems.

The common mismatch is that existing approaches often optimize or evaluate behavior within R, while the failure arises from the projection P that determines which distinctions R preserves. They can therefore improve performance inside a representation without enforcing correctness relative to the projection that produced it.

Here R should not be understood narrowly as the prompt alone. It includes whatever representational state is available to the reasoning process, including represented priors, retrieved evidence, metadata, provenance, uncertainty markers, task constraints, and explicit assumptions. The relevant question is not whether a distinction was absent from the prompt, but whether it was available, boundedly reconstructable, or explicitly assumed in the representation actually used.

8.1 Optimization Without Structural Constraints

Many reasoning systems are trained or tuned to optimize performance against specific objectives or metrics. These objectives often serve as proxies for desired behavior, but optimization against them does not by itself enforce the structural conditions required for stable reasoning under projection.

As a result, optimization can produce outputs that score highly according to the metric while violating the underlying structure of the problem. In particular:

- distinctions between competing factors may be collapsed to improve comparability,
- uncertainty may be suppressed to produce more decisive outputs,
- intermediate inconsistencies may be ignored if the final result aligns with the objective.

These effects are not anomalies. They are a consequence of optimizing over representations that do not preserve loss-relevant structure, consistent with known failures of metric-based optimization [5]. Without explicit structural constraints, optimization may favor outputs that appear correct within the metric, even when the reasoning process is unstable.

8.2 Prompting Without Dependency-Status Enforcement

In interactive systems, prompting strategies are often used to guide reasoning behavior. These strategies can improve performance in specific cases by encouraging step-by-step reasoning, reflection, or self-correction.

However, these improvements are not guaranteed to be stable. Many prompting strategies operate primarily at the level of elicited behavior and do not by themselves require explicit representation of structural loss. Such prompting alone does not generally ensure that the system:

- distinguishes between error and loss,
- maintains observability of missing information,
- preserves separation between competing interpretations.

As a result, prompting can produce reasoning that appears more detailed or cautious, while still leaving these underlying failure modes unaddressed. Improvements may degrade under slight changes in phrasing, context, or problem complexity.

Without explicit structural constraints, prompting may remain sensitive to surface variation and may not reliably preserve the conditions required for stable reasoning.

8.3 Evaluation Without Loss Visibility

Many evaluation frameworks assess correctness, consistency, or alignment with expected outputs. While these metrics are useful, they usually operate on outputs expressed in R, while the failure may arise from distinctions in S that have become indistinguishable under P.

In particular, evaluation may not detect:

- whether multiple valid interpretations were prematurely collapsed,
- whether missing information was treated as known,
- whether a conclusion was selected despite indistinguishability.

A system may produce a correct answer for the wrong reasons, or an incorrect answer that appears well-justified. In both cases, evaluation based solely on outcomes may fail to capture the underlying structural failure.

Without explicit representation of dependency status or representational loss, evaluation over outputs in R may fail to distinguish reliably between:

- correct reasoning,
- incorrect reasoning that happens to produce the correct result,
- reasoning that produces an answer where no determinate answer exists.

8.4 Structural Mismatch

Across optimization, prompting, and evaluation, a common risk emerges. These approaches operate on representations of reasoning behavior, but do not enforce the structural constraints required by reasoning under projection.

This creates a mismatch:

- reasoning requires preservation of distinctions introduced by projection,
- existing methods optimize or evaluate outputs without guaranteeing that these distinctions are maintained.

As a result, systems can appear to improve while remaining vulnerable to the same underlying failures.

8.5 Consequence

The persistence of these failure modes does not indicate that reasoning systems are fundamentally incapable. It indicates that they are being evaluated and optimized without regard to the structural conditions imposed by projection.

Unless these conditions are made explicit, improvements in scale, data, or optimization may continue to produce systems that are more capable without becoming proportionally more reliable under projection.

9 Operational Uses of Dependency Status

Although this paper does not propose a specific implementation or empirical benchmark, the dependency-status ladder suggests several operational uses. Claims can be evaluated, scaffolded, and audited according to whether their required support is preserved, reconstructable, traceable, or opaque. The structural requirements identified above function as design constraints for maintaining that status when reasoning proceeds over compressed representations.

9.1 Evaluation by Structural Shape

Evaluation can be organized around whether a response preserves the structural shape required by the query, rather than only whether it matches a domain-specific target answer. For example, a responsibility query can be evaluated for preservation of individual, process, and system levels; a high-stakes decision can be evaluated for visibility of irreversible tradeoffs and residual risk; a source-grounded answer can be evaluated for separation between observed support and plausible reconstruction; and a compositional policy problem can be evaluated for whether local conclusions are revalidated under interaction or scale.

This kind of evaluation does not require that the same domain-specific answer be expected across tasks. Instead, it asks whether the answer maintains the dependency status required by the query. In this sense, dependency status generates task-independent diagnostic questions: what distinction is needed, whether it is preserved or reconstructable, whether loss is visible, and whether the system has confused inference with decision. Such diagnostics evaluate whether the answer has the right admissibility structure, not whether the same answer is correct across all domains.

9.2 Reasoning Scaffolds and Constraint Activation

The same dependency-status distinctions can be used to constrain generation. A reasoning scaffold can require the system to preserve observability of loss-relevant state, avoid collapse of conceptual levels, keep correction bounded, distinguish error from loss, and revalidate conclusions under scale or composition. Such scaffolds do not prescribe a single reasoning style. They constrain which transformations are admissible when the system moves from a projected representation to an answer.

In addition, different tasks may require attention to different dependency-status risks. A source-grounding task may foreground observability and provenance; a responsibility-assignment task may foreground non-collapse of levels; a deployment decision may foreground bounded correction and scale revalidation. Task-specific activation of these dependency-status risks provides a control layer over reasoning without changing the underlying structural requirements.

9.3 Interface and Question-Formation Uses

Dependency-status distinctions can also be used before answer generation, as interaction lenses for forming better questions. A user interface need not treat the query as fixed. It can help users select objects, group them, specify relations, and choose a reasoning lens such as observability, reconstructability, traceability, admissibility, scale, or non-collapse. The resulting query is then not merely a request for an answer, but an explicit statement of which distinctions must be preserved during reasoning.

On this view, question formation is itself a projection-management task. The user and system jointly determine which distinctions matter before inference proceeds. This is especially important in settings where the initial question collapses roles, scales, sources, or decision criteria that must remain separate for the answer to be admissible.

9.4 Audit and Dependency Status

Finally, the dependency-status ladder suggests audit fields for downstream systems. A claim can be marked according to whether its required dependency is preserved, reconstructable, traceable, or opaque. Such markings do not guarantee correctness, but they make visible whether a conclusion is supported by the representation, recovered through an explicit path, dependent on a recorded but unrecoverable relation, or unsupported by observable provenance.

This allows downstream users or systems to distinguish claims that are inferentially supported from claims that are merely decisions, assumptions, or unresolved dependencies under non-identifiability.

These uses do not constitute a complete architecture or benchmark suite. They indicate how the structural account can guide downstream scaffolds, evaluation harnesses, runtime systems, interfaces, and audit mechanisms.

10 Scope and Limits

The framework presented in this paper describes structural conditions for reasoning under projection. It does not attempt to provide a complete theory of reasoning, nor does it claim to resolve all sources of uncertainty or failure. This section clarifies the scope of the claims and the limits of what these constraints guarantee.

10.1 What This Does Not Solve

Respecting the constraints described in this paper does not eliminate the fundamental challenges associated with reasoning over incomplete or compressed representations.

In particular, it does not:

- guarantee that a system will produce correct conclusions,
- eliminate ambiguity in cases where multiple interpretations remain consistent,
- recover distinctions that are not preserved or boundedly recoverable under the current representation,
- ensure that all relevant variables are present in the representation,
- prevent all forms of bias or misalignment in decision-making.

These limitations are not defects of the framework. They follow directly from the constraints imposed by projection. Any system operating over a reduced representation must accept that some distinctions cannot be made and some questions cannot be resolved.

10.2 Dependence on Representation

The behavior of a reasoning system under this framework depends on the representation over which it operates. Different representations preserve different distinctions, and therefore introduce different forms of loss.

Improving reasoning performance may require modifying the representation itself, not just the reasoning process applied to it. However, such changes alter the projection and therefore change the conditions under which reasoning operates.

This paper does not prescribe specific representations. It provides constraints that apply once a representation has been chosen.

10.3 Mechanism and Evaluation as Downstream Work

This paper deliberately stops at the level of structural criteria. It does not propose a complete detector for non-identifiability, a dependency-status implementation, or an empirical benchmark suite. That boundary is intentional: before such mechanisms can be evaluated, the target failure mode must be specified. The present account defines that target as the misclassification of non-identifiability under projection as recoverable inference error.

This does not mean that mechanism and evaluation are optional. It means that they are downstream tasks constrained by the structural account. A detector for representational loss would need to identify when a query depends on distinctions that are neither preserved, nor boundedly reconstructable, nor explicitly assumed in the current representation. A dependency-status implementation would need to represent whether a claim is preserved, reconstructable, traceable, or opaque. A benchmark would need to test whether a system distinguishes unsupported determinacy from ordinary error, changes behavior when the representation is enriched, and avoids presenting proxy success as target support. One natural benchmark pattern is to compare behavior under paired compressed and enriched representations, testing whether the system distinguishes unsupported determinacy from ordinary error when the compressed representation does not preserve the distinctions required by the query.

The contribution here is therefore not a complete operational solution, but a constraint that any such solution must satisfy. Mechanisms for detecting, representing, and benchmarking representational loss depend on the architecture, representation, and evaluation setting. The present paper defines the structural failure those mechanisms would need to address.

10.4 Limits of Observability

While the framework requires observability of loss-relevant state, this observability is itself subject to projection. A system may be unable to represent certain forms of loss because the representation does not include the necessary structure.

This creates an important limitation: loss can only be detected relative to distinctions that the representation is capable of expressing. A system cannot observe every loss introduced by its own projection; it can only mark the boundaries where the current representation no longer supports a requested query.

In such cases, the system may be unable to indicate the specific lost distinction, even if loss exists. It may still be able to mark a weaker diagnostic condition, such as missing variables, unsupported assumptions, or insufficient provenance. This places a practical limit on how completely loss can be made visible within any given representation.

As a result, the framework supports improvement of observability, but does not guarantee its completeness.

11 Conclusion

Reasoning systems operate over representations that compress the underlying structure of the problems they address. This compression is unavoidable, but it can collapse distinctions that later reasoning requires.

The central claim of this paper is that a recurring class of reasoning failures arises when systems treat such loss as recoverable error. When a query is not identifiable under the current representation, additional inference over that representation cannot recover the missing distinction. A determinate answer may still be chosen, but it must be marked as a decision, assumption, conditional claim, or representation expansion rather than as an inference determined by the original representation.

The error–loss distinction motivates a dependency-status ladder: claims may depend on distinctions that are preserved, reconstructable, traceable, or opaque after projection. The structural requirements of observability, bounded correction, and non-collapse of distinct representational levels are constraints for preserving and acting on that status during reasoning. These requirements do not guarantee correctness. They constrain the relation between claims and the representations that support them.

The broader implication is that improvements in scale, optimization, model complexity, or inference quality can increase performance over a representation without restoring distinctions that the projection failed to preserve. A system may become more capable, coherent, and persuasive while remaining unable to justify certain conclusions from the information available to it.

Reasoning under projection is therefore not a rare edge case. It is the ordinary condition of complex reasoning over compressed, partial, or task-shaped representations. The question is not whether systems will reason under projection, but whether they can reliably distinguish what their representations preserve, what they can reconstruct, what they can merely trace, and what they have lost.

References

[1] Dario Amodei et al. Concrete Problems in AI Safety. arXiv:1606.06565, 2016.

[2] David Blackwell. Equivalent comparisons of experiments. The Annals of Mathematical Statistics, 1953.

[3] Edmund M. Clarke, Orna Grumberg, Somesh Jha, Yuan Lu, and Helmut Veith. Counterexample-guided abstraction refinement. CAV, 2000.

[4] Patrick Cousot and Radhia Cousot. Abstract interpretation: A unified lattice model for static analysis of programs by construction or approximation of fixpoints. POPL, 1977.

[5] Charles Goodhart. Problems of Monetary Management: The U.K. Experience. 1975.

[6] Leslie Pack Kaelbling, Michael L. Littman, and Anthony R. Cassandra. Planning and acting in partially observable stochastic domains. Artificial Intelligence, 1998.

[7] Lihong Li, Thomas J. Walsh, and Michael L. Littman. Towards a unified theory of state abstraction for MDPs. ISAIM, 2006.

[8] Judea Pearl. Causality: Models, Reasoning, and Inference. Cambridge University Press, 2009.

[9] Leonard J. Savage. The Foundations of Statistics. 1954.

[10] Claude Shannon. A Mathematical Theory of Communication. Bell System Technical Journal, 1948.

[11] Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. 1999.